Lecture 39

Triple integrals and moments

Uses.

Summary 8: Scalar product and chain rules differentiate coordinate formulas.
Summary 9: Derivative signs determine increase, decrease and interior extrema.
Summary 12: Euclidean scalar products and norms describe lengths and perpendicular projections.
Summary 21: The absolute determinant gives the volume factor of a linear coordinate map.
Summary 26: Proper continuous integrals obey substitution, parts and the fundamental theorem.
Summary 27: Ordinary improper integrals require separate endpoint limits. For $s > 0$, $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,\mathrm{d}t$, with $\Gamma(s+1) = s\Gamma(s)$.
Summary 36: The Jacobian is the derivative matrix, and compositions obey the matrix chain rule.
Summary 38: Standing fact M gives proper compact Fubini and admissible changes of variables; $\Gamma(1/2) = \sqrt{\pi}$.

Objectives. *A solid has a mass distribution over volume rather than area. Triple integrals evaluate its mass and moments; cylindrical and spherical coordinates make symmetry useful without losing the volume factor. We derive the moment of inertia of a uniform ball and then use ordinary radial limits to normalise a Gaussian distribution in three dimensions.*

By the end of this lecture you will be able to:

- Describe a compact solid by iterated bounds and check the hypotheses of standing fact M.
- Derive the cylindrical and spherical Jacobians with the stated angle conventions.
- Compute mass, centroid and moments of inertia, and prove the parallel-axis relation.
- Normalise a radial Gaussian and derive its mean, mean square and most frequent radius using ordinary convergent integrals.

39.1 Volume integrals on compact solids

A small-volume mass sum has the same form as the small-area sum of Lecture 38, with cell volume in place of area. The triple Riemann integral is its common limit as the largest cell diameter tends to zero, independently of the selected sample points:

$$\iiint_K f\,\mathrm{d}V = \lim \sum_j f(\xi_j)\,\mathrm{volume}(K_j).$$

In particular, integrating 1 gives volume. Finite linearity, order and additivity follow from the corresponding properties of the finite sums. For nonnegative f, inclusion of compact regions preserves integral order when the closure of their difference is a finite union of admissible graph solids, by additivity and positivity as in the plane.

Here a compact graph solid has a compact planar graph region D and bounds

$$K = \{(x, y, z) : (x, y) \in D,\ g_1(x, y) \le z \le g_2(x, y)\},$$

where $g_1 \le g_2$ are continuous. Coordinate permutations and finite unions of such pieces are allowed. The boundary is piecewise C^1 in this sense: it is covered by finitely many C^1 parametrised surface pieces, with parametrisations C^1 up to their parameter boundaries. Away from joining curves and finitely many joining points, each piece is locally a C^1 graph of one coordinate as a function of the other two. Graph regularity may fail at those joining locations, allowing, for example, a cone vertex. All regions used here are closed and bounded, and their boundaries have this property. These are the three-dimensional graph and boundary conditions in the standing fact M of Lecture 38.

For continuous f on such a solid, M gives existence and compact Fubini:

$$\iiint_K f \,\mathrm{d}V = \iint_D \left(\int_{g_1(x,y)}^{g_2(x,y)} f(x, y, z) \,\mathrm{d}z \right) \mathrm{d}A.$$

For each integration, variables not yet integrated are held fixed. Every change of order must describe the same solid; projecting its bounds may require a finite division into graph pieces. No unbounded solid or singular integrand has yet been admitted.

For a continuous volume density $\rho \ge 0$ on K, define mass and first coordinate moments by

$$m = \iiint_K \rho \,\mathrm{d}V, \qquad Q_x = \iiint_K x\rho \,\mathrm{d}V, \quad Q_y = \iiint_K y\rho \,\mathrm{d}V, \quad Q_z = \iiint_K z\rho \,\mathrm{d}V.$$

If $m > 0$, its centroid is $\boldsymbol{c} = (Q_x, Q_y, Q_z)/m$. Thus the planar weighted-coordinate definition extends by replacing area with volume and including the third coordinate.

Example 39.1 (A tetrahedron in Cartesian coordinates)**.** Let $K = \{x, y, z \ge 0 : x + y + z \le 1\}$. It is a compact graph solid with planar triangle $0 \le x \le 1$, $0 \le y \le 1 - x$ and height $1 - x - y$. Its planar faces are C^1. Its volume is

$$\iiint_K 1 \,\mathrm{d}V = \int_0^1 \int_0^{1-x} (1 - x - y) \,\mathrm{d}y \,\mathrm{d}x = \int_0^1 \frac{(1-x)^2}{2} \,\mathrm{d}x = \frac{1}{6}.$$

For uniform unit density, its first z moment is

$$Q_z = \int_0^1 \int_0^{1-x} \frac{(1 - x - y)^2}{2} \,\mathrm{d}y \,\mathrm{d}x = \int_0^1 \frac{(1-x)^3}{6} \,\mathrm{d}x = \frac{1}{24}.$$

Permuting x, y, z preserves the solid and has absolute Jacobian 1. Thus all three first moments agree, giving centroid $(1/4, 1/4, 1/4)$. This use of symmetry is a change of variables on the actual region.

39.2 Cylindrical coordinates

For a change $(x, y, z) = \Phi(u, v, w)$ satisfying M, the volume rule is

$$\iiint_K f(x, y, z) \,\mathrm{d}V = \iiint_E f(\Phi(u, v, w)) \,|\det J_\Phi| \,\mathrm{d}u \,\mathrm{d}v \,\mathrm{d}w.$$

The map must be C^1 on a neighbourhood of the compact parameter solid, onto the physical solid, injective and nonsingular on the parameter interior. The boundary exceptions in M include angular seams, axes and the origin. The determinant is taken in the displayed order of the parameter coordinates. Its absolute value, rather than its sign, measures volume.

Cylindrical coordinates retain the Cartesian height and use plane polar coordinates horizontally:

$$x = r\cos\varphi, \qquad y = r\sin\varphi, \qquad z = z, \qquad r \geq 0.$$

The reuse of z means the height is unchanged. In the parameter order (r, φ, z) the Jacobian is

$$J_\Phi = \begin{pmatrix} \cos\varphi & -r\sin\varphi & 0 \\ \sin\varphi & r\cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det J_\Phi = r.$$

Expansion along the last row reduces this determinant to the plane polar one. A single angular sweep of width at most 2π is injective in the interior when $r > 0$. Hence $\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\varphi\,\mathrm{d}z$. Taking an angular interval twice as long would double-count ordinary interior points and violate M.

Example 39.2 (Mass below a paraboloid)**.** Consider the compact solid and continuous density

$$K = \{x^2 + y^2 \leq 1,\ 0 \leq z \leq 1 - x^2 - y^2\}, \qquad \rho(x, y, z) = 1 + z.$$

Its boundary consists of a smooth paraboloid cap and a flat disc meeting along a circle. Cylindrical bounds are $0 \leq r \leq 1$, $0 \leq \varphi \leq 2\pi$, $0 \leq z \leq 1 - r^2$. Writing $h = 1 - r^2$, integration first in height gives

$$m = 2\pi \int_0^1 \left(h + \frac{h^2}{2}\right) r\,\mathrm{d}r = \pi \int_0^1 \left(h + \frac{h^2}{2}\right) \mathrm{d}h = \frac{2\pi}{3}.$$

The substitution reverses the radial endpoints and contributes $\mathrm{d}h = -2r\,\mathrm{d}r$. The first height moment is similarly

$$Q_z = 2\pi \int_0^1 \left(\frac{h^2}{2} + \frac{h^3}{3}\right) r\,\mathrm{d}r = \pi\left(\frac{1}{6} + \frac{1}{12}\right) = \frac{\pi}{4}.$$

Reflection in either vertical coordinate plane preserves both K and its density. Substituting $x \mapsto -x$ reverses the sign of Q_x while leaving its value unchanged, so $Q_x = 0$; the same argument gives $Q_y = 0$. Consequently the centroid is $(0, 0, 3/8)$. The density factor was retained in both the mass and the first moment; normalising a geometric volume moment would describe a different body.

39.3 Spherical coordinates and their volume factor

Use r for distance from the origin, θ for the polar angle measured from the positive z axis, and φ for the azimuth in the xy plane. This spherical r is the distance denoted ρ in Lecture 14; it is not the cylindrical r (distance from the z axis) in the preceding section. We use ρ below for mass density, so the present section keeps r for the spherical radius. The coordinate map is

$$x = r\sin\theta\cos\varphi, \quad y = r\sin\theta\sin\varphi, \quad z = r\cos\theta, \qquad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi \leq 2\pi.$$

Figure 39.1 fixes the angle conventions before the Jacobian is computed.

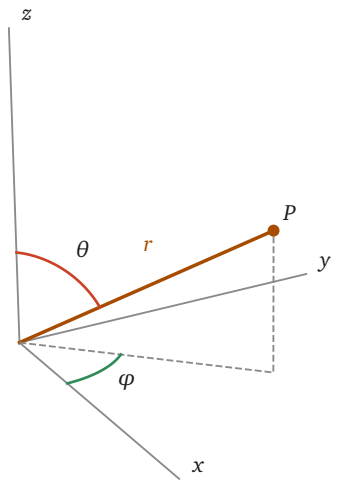


Figure 39.1: The radius joins the origin to P. The polar angle θ starts at the positive z axis; the azimuth φ starts at the positive x axis in the horizontal plane.

The two angles have different roles. In particular, $\theta = 0$ is the positive axis, while $\theta = \pi/2$ is the equatorial plane. In the parameter order (r, θ, φ),

$$J_\Phi = \begin{pmatrix} \sin\theta\cos\varphi & r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ \sin\theta\sin\varphi & r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix}.$$

Factor r from the second column and $r\sin\theta$ from the third. Expansion along the last row of the remaining matrix gives $\cos^2\theta + \sin^2\theta = 1$. Thus

$$\det J_\Phi = r^2\sin\theta, \qquad \mathrm{d}V = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi.$$

For positive radius, $0 < \theta < \pi$ and $0 < \varphi < 2\pi$ the map is injective with positive determinant. The axis, origin and angular seam are parameter-boundary exceptions allowed by M. For the ball $B_R = \{x^2 + y^2 + z^2 \le R^2\}$, $R > 0$, take $0 \le r \le R$. The map is smooth even on a neighbourhood of this parameter box. The sphere has finitely many smooth graph pieces: on a piece where one coordinate has a fixed sign and absolute value at least $R/\sqrt{3}$, solve for that coordinate as the corresponding signed square root. These six pieces cover the sphere and satisfy the boundary condition.

39.4 Mass and the moment of inertia of a ball

For a line through the origin with unit direction $\boldsymbol{n}$, the squared perpendicular distance of $\boldsymbol{x} = (x, y, z)$ from the line is $\|\boldsymbol{x}\|^2 - (\boldsymbol{n}\cdot\boldsymbol{x})^2$, by perpendicular projection. The *moment of inertia* about that axis is therefore

$$I_{\boldsymbol{n}} = \iiint_K \rho(\boldsymbol{x})\left(\|\boldsymbol{x}\|^2 - (\boldsymbol{n}\cdot\boldsymbol{x})^2\right)\mathrm{d}V.$$

It measures how far mass lies from an axis, not simply from one point. For the z axis the squared distance is $x^2 + y^2$.

For uniform positive density ρ_0 on B_R, reflection symmetry gives centroid zero. Spherical integration yields

$$m = \rho_0 \int_0^R r^2 \,\mathrm{d}r \int_0^\pi \sin\theta \,\mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\varphi = \frac{4\pi\rho_0 R^3}{3}.$$

The z-axis squared distance is $r^2 \sin^2\theta$, so

$$I_z = 2\pi\rho_0 \int_0^R r^4 \,\mathrm{d}r \int_0^\pi \sin^3\theta \,\mathrm{d}\theta = 2\pi\rho_0 \frac{R^5}{5}\frac{4}{3} = \frac{8\pi\rho_0 R^5}{15} = \frac{2}{5}mR^2.$$

For the angular integral put $u = \cos\theta$ to obtain $\int_{-1}^{1}(1-u^2)\,\mathrm{d}u = 4/3$. The two factors of $\sin\theta$ from squared distance are additional to the one in the volume element.

This value holds for every central axis, which can also be verified without constructing a rotation. Each mixed moment, such as $\iiint_{B_R} \rho_0 xy \,\mathrm{d}V$, vanishes by reflection in one coordinate. Permuting coordinates shows that the three diagonal second moments agree; call their common value A. For $\|\boldsymbol{n}\| = 1$, expanding the square gives

$$\iiint_{B_R} \rho_0 (\boldsymbol{n}\cdot\boldsymbol{x})^2 \,\mathrm{d}V = (n_1^2 + n_2^2 + n_3^2)A = A, \qquad I_{\boldsymbol{n}} = 3A - A = 2A = I_z.$$

All symmetry substitutions are linear bijections of the same ball, with absolute determinant 1.

39.5 The parallel-axis relation

The preceding definition extends to any line $\ell = \{\boldsymbol{a} + t\boldsymbol{n} : t \in \mathbb{R}\}$ with $\|\boldsymbol{n}\| = 1$:

$$I_\ell = \iiint_K \rho(\boldsymbol{x}) \big(\|\boldsymbol{x} - \boldsymbol{a}\|^2 - (\boldsymbol{n}\cdot(\boldsymbol{x}-\boldsymbol{a}))^2 \big) \,\mathrm{d}V.$$

Let $\boldsymbol{c}$ be the centroid and put $\boldsymbol{u} = \boldsymbol{x} - \boldsymbol{c}$, $\boldsymbol{d} = \boldsymbol{c} - \boldsymbol{a}$. Expanding both squares gives the pointwise identity

$$\begin{aligned} \|\boldsymbol{u}+\boldsymbol{d}\|^2 - (\boldsymbol{n}\cdot(\boldsymbol{u}+\boldsymbol{d}))^2 = {} & \|\boldsymbol{u}\|^2 - (\boldsymbol{n}\cdot\boldsymbol{u})^2 + \|\boldsymbol{d}\|^2 - (\boldsymbol{n}\cdot\boldsymbol{d})^2 \\ & + 2\big(\boldsymbol{u}\cdot\boldsymbol{d} - (\boldsymbol{n}\cdot\boldsymbol{u})(\boldsymbol{n}\cdot\boldsymbol{d})\big). \end{aligned}$$

By the centroid definition, $\iiint_K \rho\boldsymbol{u}\,\mathrm{d}V = \boldsymbol{0}$. The integral of the entire cross term is consequently zero. The constant term integrates to its value times m. Writing $\ell_c = \{\boldsymbol{c} + t\boldsymbol{n} : t \in \mathbb{R}\}$ and $d_\perp$ for the perpendicular distance between these two parallel lines, we obtain

$$I_\ell = I_{\ell_c} + m d_\perp^2, \qquad d_\perp^2 = \|\boldsymbol{c}-\boldsymbol{a}\|^2 - (\boldsymbol{n}\cdot(\boldsymbol{c}-\boldsymbol{a}))^2. \tag{39.1}$$

This is the parallel-axis theorem [P]. It applies to a continuous nonnegative density with positive mass on an admissible compact solid. The reference axis must pass through the centroid for this form of the formula; otherwise the discarded cross term need not vanish. For a uniform ball, an axis at distance $b \ge 0$ from the centre has $I = m(2R^2/5 + b^2)$. A tangent axis has $b = R$, giving $I = 7mR^2/5$. This computes an offset axis directly from the central calculation.

39.6 Gaussian radial mass and Gamma moments

Let $a > 0$. A radial density has the form $\rho(\boldsymbol{x}) = h(\|\boldsymbol{x}\|)$. Spherical integration on each compact ball gives

$$\iiint_{B_R} h(\|\boldsymbol{x}\|)\,\mathrm{d}V = 4\pi \int_0^R h(r) r^2\,\mathrm{d}r$$

when h is continuous on $[0, R]$. We define its full radial mass by the ordinary one-variable limit of the right side as $R \to \infty$, when this limit is finite. This definition does not use an improper version of M. For nonnegative density, the compact masses increase with radius, so the limit cannot depend on cancellation.

For the Gaussian model choose

$$\rho_a(\boldsymbol{x}) = \left(\frac{a}{\pi}\right)^{3/2} e^{-a\|\boldsymbol{x}\|^2}, \qquad w_a(r) = \frac{4a^{3/2}}{\sqrt{\pi}} r^2 e^{-ar^2} \quad (r \geq 0).$$

The factor w_a is mass per unit radius after the angular coordinates have been integrated; it differs from the density per unit volume. With $t = ar^2$, ordinary substitution on finite positive cutoffs and then the endpoint limits gives, for $k \geq 0$,

$$\int_0^\infty r^k w_a(r)\,\mathrm{d}r = \frac{2}{\sqrt{\pi}} a^{-k/2} \int_0^\infty t^{(k+3)/2-1} e^{-t}\,\mathrm{d}t = \frac{2}{\sqrt{\pi}} a^{-k/2} \Gamma\left(\frac{k+3}{2}\right). \tag{39.2}$$

The Gamma parameter is positive, so convergence at both endpoints follows from Lecture 27. At $k = 0$, r^0 means the constant function 1, including at the origin. The formula and $\Gamma(3/2) = \sqrt{\pi}/2$ show that total mass is 1. Thus w_a is a normalised nonnegative density on the half-line.

For clarity, the half-integer Gamma values used here all follow from a proved recurrence. Starting with $\Gamma(1/2) = \sqrt{\pi}$ and multiplying successively by $1/2, 3/2, \ldots, n - 1/2$ gives

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{1 \cdot 3 \cdots (2n-1)}{2^n} \sqrt{\pi} = \frac{(2n)!}{4^n n!} \sqrt{\pi} \qquad (n = 0, 1, 2, \ldots).$$

For $n = 0$ the product is empty and equals 1. The factorial expression follows by splitting $(2n)!$ into its even and odd factors. In particular $\Gamma(3/2) = \sqrt{\pi}/2$ and $\Gamma(5/2) = 3\sqrt{\pi}/4$. Also $\Gamma(2) = 1$ by the integer recurrence. Substitution in Eq. (39.2) yields the mean radius, mean square radius and root mean square radius:

$$\bar{r} = \frac{2}{\sqrt{\pi a}}, \qquad \overline{r^2} = \frac{3}{2a}, \qquad r_{\mathrm{rms}} = \sqrt{\frac{3}{2a}}.$$

These are different operations on the same density. To find where the radial density is largest, differentiate it for $r > 0$:

$$w_a'(r) = \frac{8a^{3/2}}{\sqrt{\pi}} r e^{-ar^2}(1 - ar^2).$$

It is positive before $r = 1/\sqrt{a}$ and negative afterwards. Since $w_a(0) = 0$ and $w_a(r) > 0$ for $r > 0$, the unique global maximum occurs at $r_{\mathrm{mode}} = 1/\sqrt{a}$.

Figure 39.2 shows how the radial profile changes with the positive parameter.

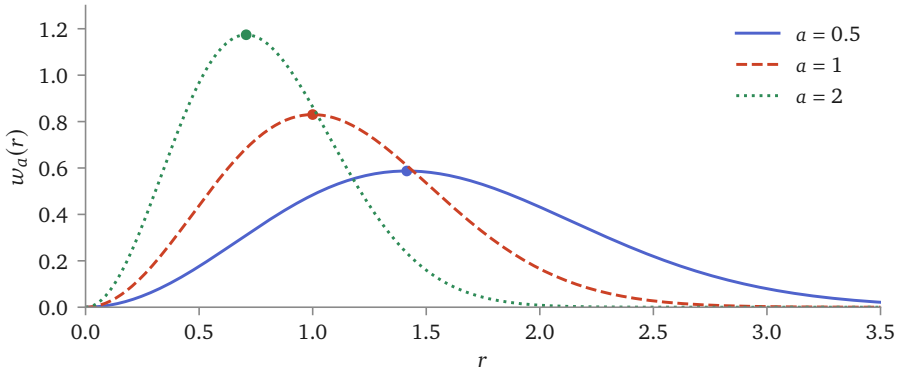


Figure 39.2: Normalised radial densities for three positive parameters. The marked maximum of each curve occurs at $r = 1/\sqrt{a}$; the vertical coordinate is mass per unit radius.

The volume density, by contrast, is largest at the origin; multiplication by the spherical shell factor changes the question being answered.

In the Maxwell speed model, the positive physical constants are particle mass m_0, temperature T and Boltzmann constant k_B, and the model parameter is $a = m_0/(2k_BT)$. Taking r to denote speed gives the three characteristic values

$$r_{\text{mode}} = \sqrt{\frac{2k_BT}{m_0}}, \qquad \bar{r} = \sqrt{\frac{8k_BT}{\pi m_0}}, \qquad r_{\text{rms}} = \sqrt{\frac{3k_BT}{m_0}}.$$

The Gaussian density is an input of this physical model. The normalisation and moments just computed are mathematical consequences of that input, rather than a derivation of the model from mechanics.

Summary 39.

- Standing fact M applies to continuous integrands on the stated compact graph solids under the piecewise C^1 boundary convention, including its allowed joining locations. Iterated bounds must describe the same solid.
- Cylindrical coordinates have volume factor r. With polar angle θ measured from the positive z axis, spherical coordinates have factor $r^2 \sin\theta$.
- A uniform ball has moment $2mR^2/5$ about any central axis. For a continuous density of positive mass, Eq. (39.1) transfers a centroidal moment to a parallel axis.
- The radial Gaussian moments Eq. (39.2) follow by ordinary endpoint limits and the Gamma integral. The normalised radial density has mode $1/\sqrt{a}$, mean $2/\sqrt{\pi a}$ and mean square $3/(2a)$ for $a > 0$.

Sheet 39 develops other solids, radial powers, the Beta–Gamma relation and elementary Laplace transforms with their convergence domains. Lecture 40 integrates vector fields along curves and asks when work depends only on the endpoints.

References

V. A. Zorich, *Mathematical Analysis II*, 2nd English ed., Springer, 2016, Chapter 11 (compact multiple integrals and changes of variables). J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§15.6–15.9.